

1 Different species of animal have neurones with different characteristics.

(a) Fig. 1.1 is a diagram of a motor neurone of a rat and a motor neurone of a snail.

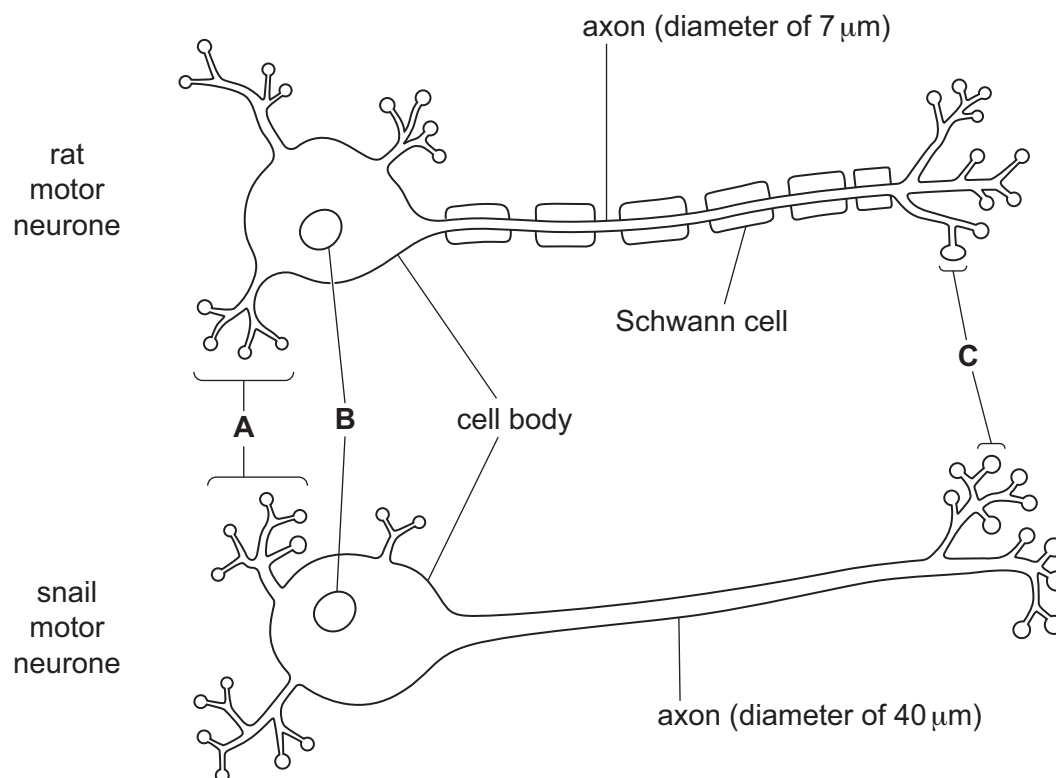


Fig. 1.1

(i) Name the structures labelled **A**, **B** and **C** on Fig. 1.1.

A

B

C [3]

(ii) The rat motor neurone has an impulse transmission speed of 50 ms^{-1} . The snail motor neurone has an impulse transmission speed of 8 ms^{-1} .

Explain why the rat motor neurone has a faster impulse transmission speed than the snail motor neurone.

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..... [2]

- (b) Fig. 1.2 shows an action potential in a rat neurone and Fig. 1.3 shows an action potential in a snail neurone.

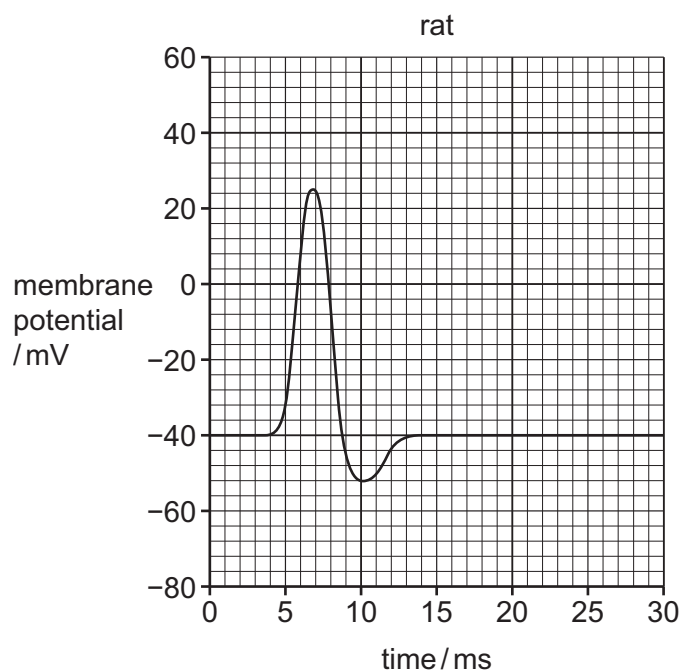


Fig. 1.2

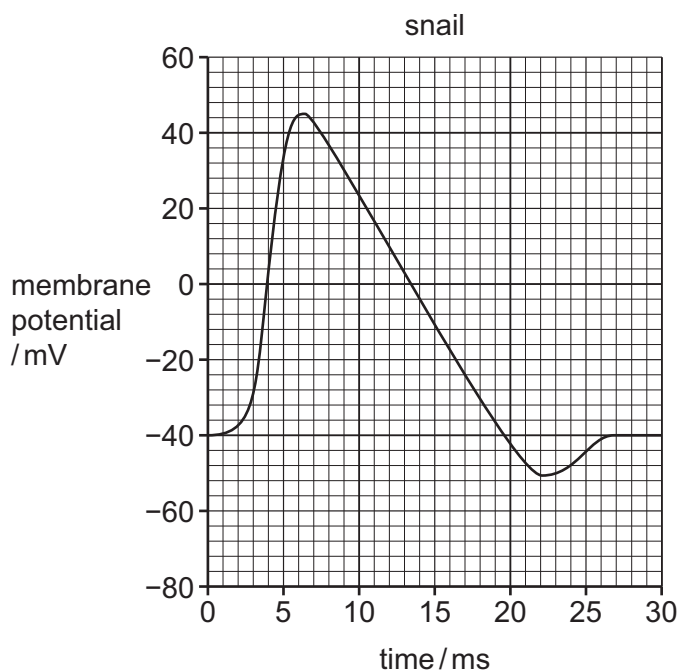


Fig. 1.3

Contrast the two action potentials shown in Fig. 1.2 and Fig. 1.3.

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..... [3]

[Total: 8]